Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Is a ray infinite in one direction? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **A ray has one endpoint and extends forever in one direction. The answer is yes.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Does a line extend infinitely in both directions? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **A line has no endpoints and continues forever in both directions.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

How many points are needed to define a unique line? Student's answer: 1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: Two distinct points determine exactly one line.

Correct answer: 2

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A reflection maps segment AB to A′B′. Is AB congruent to A′B′? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Reflections are rigid motions. They preserve length, so the segments are congruent.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A 40° angle is rotated about a point. What is the measure of the image angle? Student's answer: 80°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Rotations preserve angle measure. The image angle is still 40°.**

Correct answer: **40°**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Does a dilation preserve distances? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **A dilation with scale factor not equal to 1 changes distances, so it is not a rigid motion.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$\triangle ABC \cong \triangle DEF$, and $\angle A = 50°$. What is $\angle D$? Student's answer: 130°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Corresponding angles of congruent triangles are congruent. $\angle D = 50°$.**

Correct answer: **50°**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$\triangle PQR \cong \triangle XYZ$, and PQ = 7. What is XY? Student's answer: 14

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student doubled the length. Corresponding sides are equal, so XY = 7.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In $\triangle ABC \cong \triangle DEF$ with A ↔ D, B ↔ E, and C ↔ F, which side corresponds to AC? Student's answer: DE

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **AC uses the vertices A and C, which match D and F. The corresponding side is DF.**

Correct answer: DF

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Two triangles have sides 3, 4, and 5 in both. Are they congruent? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **All three pairs of sides are congruent, so the triangles are congruent by SSS.**

Correct answer: Yes, by SSS

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Two triangles have angles 50° and 60° with a side of 4 between those angles in both. Which criterion proves congruence? Student's answer: AAA

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The side is between the two angles, so the criterion is ASA. AAA does not prove congruence.**

Correct answer: **ASA**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Is SSA a valid criterion for proving triangles congruent? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **SSA can describe two different triangles, so it is not a valid congruence criterion.**

Correct answer: No

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Two angles are vertical angles. One measures 70°. What is the other? Student's answer: 110°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Vertical angles are congruent. The other angle is 70°.**

Correct answer: **70°**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Two parallel lines are cut by a transversal. An alternate interior angle is 65°. What is the other?

Student's answer: 115°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Alternate interior angles are congruent, so the other angle is 65°.**

Correct answer: 65°

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A point is on the perpendicular bisector of a segment. Is it equidistant from the endpoints?

Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Every point on the perpendicular bisector is equidistant from the endpoints.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A triangle has angles of 40° and 75°. What is the third angle? Student's answer: 75°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student copied one angle. 180 − (40 + 75) = 65°.**

Correct answer: 65°

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A triangle has a base of 12. What is the length of its midsegment parallel to the base? Student's answer: 12

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The midsegment parallel to the third side is half that side. Its length is 6.**

Correct answer: 6

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

An exterior angle of a triangle is 115°. One remote interior angle is 50°. What is the other remote interior angle? Student's answer: 165°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the angles. 115 - 50 = 65°.**

Correct answer: **65°**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In a parallelogram, one angle is 110°. What is the opposite angle? Student's answer: 70°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Opposite angles of a parallelogram are congruent. The opposite angle is 110°.**

Correct answer: 110°

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

The diagonals of a parallelogram meet at O. If AC = 14, what is AO? Student's answer: 14

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The diagonals bisect each other, so AO is half of AC. AO = 7.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In a parallelogram, one angle is 110°. What is an adjacent angle? Student's answer: 110°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Consecutive angles of a parallelogram add to 180°. The adjacent angle is 70°.**

Correct answer: 70°

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A dilation with scale factor 3 maps a side of length 4 to a new side. What is the new length?

Student's answer: 7

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 3 to 4. Multiply: 4 × 3 = 12.**

Correct answer: **12**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Are two triangles similar if two pairs of angles are congruent? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Two pairs of congruent angles guarantee similarity by AA.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In a similar pair, a side of length 3 in the small triangle matches a side of 6 in the large triangle. A side of 4 in the small triangle is what in the large? Student's answer: 6

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The scale factor is 6 ÷ 3 = 2. So 4 × 2 = 8.**

Correct answer: **8**

Find the Error

Grade 10 · G.SRT.4

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Similar triangles have AB = 4, DE = 6, and BC = 8. What is EF if EF corresponds to BC? Student's answer: 10

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{4}{6}$ = 8/x, so x = 12. The student added the sides.

Correct answer: 12

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A line parallel to BC crosses AB at D and AC at E. AD = 3, DB = 6, and AE = 4. What is EC? Student's answer: 2

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: AD/DB = AE/EC, so $\frac{3}{6}$ = 4/EC. Then EC = 8.

Correct answer: 8

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In a right triangle, the altitude to the hypotenuse splits it into segments of 4 and 9. What is the altitude? Student's answer: 13

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The altitude is the geometric mean of the segments: $\sqrt{(4 \times 9)} = 6$.**

Correct answer: 6

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Triangles have two sides and the included angle congruent. Which method proves congruence?

Student's answer: SSA

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The angle is between the two sides, so the correct criterion is SAS.**

Correct answer: **SAS**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Similar triangles have sides 5 and 10 matching, and a side of 14 in the second triangle matches a side of x in the first. Find x. Student's answer: 9

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{5}{10}$ **= x/14, so x = 7.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In similar triangles, AB = 3 and DE = 6, and AC = 5. What is DF? Student's answer: 5

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The scale factor is 6 ÷ 3 = 2. DF = 2 × 5 = 10.**

Correct answer: **10**

Find the Error

Grade 10 · G.SRT.6

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In a right triangle, the opposite side to angle A is 3 and the hypotenuse is 5. What is sin A?

Student's answer: $\frac{4}{5}$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{4}{5}$ is cos of the other angle. sin A = $\frac{3}{5}$.

Correct answer: $\frac{3}{5}$

Find the Error

Grade 10 · G.SRT.6

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In a right triangle, the adjacent side to angle B is 8 and the hypotenuse is 17. What is cos B?

Student's answer: $\frac{15}{17}$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{15}{17}$ uses the opposite side. cos B = $\frac{8}{17}$.

Correct answer: $\frac{8}{17}$

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is tan 45°? Student's answer: $\sqrt{2}$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **In a 45-45-90 triangle, the opposite and adjacent legs are equal, so tan 45° = 1.**

Correct answer: 1

Find the Error

Grade 10 · G.SRT.8

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A right triangle has legs 3 and 4. What is the hypotenuse? Student's answer: 7

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the legs. $3^2 + 4^2 = 25$, so the hypotenuse is 5.**

Correct answer: 5

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A right triangle has opposite side 5 and hypotenuse 10 for angle A. What is angle A? Student's answer: 60°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **sin A = $\frac{5}{10}$ = $\frac{1}{2}$, so A = 30°.**

Correct answer: **30°**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A 10-foot ladder makes a 60° angle with the ground. How high does it reach? Student's answer: 5 feet

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student used cosine. Height = 10 sin 60° = $5\sqrt{3}$, about 8.66 feet.**

Correct answer: **$5\sqrt{3}$, about 8.66 feet**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A central angle measures 80°. What is the inscribed angle that intercepts the same arc?

Student's answer: 80°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **An inscribed angle is half the central angle that intercepts the same arc. 80 ÷ 2 = 40°.**

Correct answer: 40°

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

An inscribed angle measures 35°. What is the intercepted arc? Student's answer: 35°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The intercepted arc is twice the inscribed angle, so 2 × 35 = 70°.**

Correct answer: 70°

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

An inscribed angle is drawn with its sides on a diameter. What is its measure? Student's answer: 180°

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The intercepted arc is a semicircle, which is 180°. The inscribed angle is half of that, so it is 90°.**

Correct answer: 90°

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the arc length of a 90° arc in a circle with radius 6. Student's answer: 6π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student found the length of a full half circle. $\left(\frac{90}{360}\right) \times 12\pi = 3\pi$, about 9.42.**

Correct answer: **3π, about 9.42**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the area of a 60° sector in a circle with radius 3. Student's answer: 9π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student found the area of the full circle. $\left(\frac{60}{360}\right) \times 9\pi = 1.5\pi$, about 4.71.**

Correct answer: **1.5π, about 4.71**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the arc length of a 120° arc in a circle with radius 9. Student's answer: 12π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student used $2\pi \times 9 \times \frac{2}{3}$ incorrectly. $\left(\frac{120}{360}\right) \times 18\pi = 6\pi$.**

Correct answer: 6π

Find the Error

Grade 10 · G.GPE.4

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A rectangle has vertices A(0, 0), B(4, 0), C(4, 3), and D(0, 3). What is the length of diagonal AC?

Student's answer: 7

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the side lengths. AC = $\sqrt{(4^2 + 3^2)} = 5$.**

Correct answer: 5

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the slope of the line through A(1, 2) and B(5, 10)? Student's answer: $\frac{1}{2}$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student flipped the ratio. The slope is 8 ÷ 4 = 2.**

Correct answer: 2

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Are the lines y = 2x + 1 and y = 2x - 3 parallel? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Both lines have slope 2 and different intercepts, so they are parallel.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the midpoint of segment AB with A(0, 0) and B(10, 0). Student's answer: (4, 0)

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a subtraction error. The midpoint is (5, 0).**

Correct answer: **(5, 0)**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the point $\frac{1}{3}$ of the way from A(0, 0) to B(12, 0). Student's answer: (3, 0)

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided 12 by 4. $\frac{1}{3}$ of 12 is 4, so the point is (4, 0).**

Correct answer: **(4, 0)**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the point that divides A(2, 0) to B(8, 0) in a 2:1 ratio, measured from A. Student's answer: (5, 0)

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student used $\frac{1}{2}$ of the distance. $\frac{2}{3}$ of 6 is 4, so the point is (6, 0).**

Correct answer: **(6, 0)**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A rectangle has vertices (0, 0), (5, 0), (5, 2), and (0, 2). What is its perimeter? Student's answer: 10

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted each side once. The perimeter is 2(5 + 2) = 14.**

Correct answer: 14

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A right triangle has vertices (0, 0), (6, 0), and (0, 8). What is its area? Student's answer: 14

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the legs. Area = $\frac{1}{2}$ × 6 × 8 = 24.**

Correct answer: 24

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A square has vertices (1, 1), (4, 1), (4, 4), and (1, 4). What is its area? Student's answer: 12

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: 12 is the perimeter. The side is 3, so the area is 9.

Correct answer: 9

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the circumference of a circle with radius 4. Student's answer: 4π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to multiply by 2. $C = 2\pi(4) = 8\pi$, about 25.13.**

Correct answer: **8π, about 25.13**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the volume of a cylinder with radius 3 and height 5. Student's answer: 15π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to square the radius. V = π(9)(5) = 45π, about 141.37.**

Correct answer: **45π, about 141.37**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Find the volume of a cone with radius 3 and height 6. Student's answer: 54π

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot the $\frac{1}{3}$. $V = (\frac{1}{3})\pi(9)(6) = 18\pi$, about 56.55.**

Correct answer: **18π, about 56.55**

Find the Error

Grade 10 · G.GMD.3

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A box is 4 by 5 by 6 units. What is its volume? Student's answer: 15

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied only two dimensions. V = 4 × 5 × 6 = 120.**

Correct answer: **120**

Find the Error

Grade 10 · G.GMD.3

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A cylinder has volume 200π and radius 5. What is its height? Student's answer: 40

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided by 5 instead of 25. 200π = π(25)h, so h = 8.**

Correct answer: 8

Find the Error

Grade 10 · G.GMD.3

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A sphere has volume 36π. What is its radius? Student's answer: 9

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided the volume by π and stopped. $36 = \left(\frac{4}{3}\right)r^3$ gives $r^3 = 27$, so $r = 3$.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Which solid best models a soup can? Student's answer: cone

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **A soup can has two circular bases and straight sides, so a cylinder is the best model.**

Correct answer: **Cylinder**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Which solid best models a basketball? Student's answer: cylinder

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **A basketball is round in every direction, so a sphere is the best model.**

Correct answer: **Sphere**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A rectangular garden is 12 feet by 7 feet and needs fencing around its edge. How much fencing is needed? Student's answer: 19 feet

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student added only one length and one width. Perimeter = 2(12 + 7) = 38 feet.**

Correct answer: **38 feet**